CLASS 10 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): Rusting of iron is a chemical change. Reason (R): A new substance with different properties (iron oxide) is formed. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Dry HCl gas does not change the colour of dry blue litmus paper. Reason (R): HCl gas does not ionise in the absence of water to release H^+ ions. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): Sodium is stored in kerosene oil. Reason (R): Sodium reacts vigorously with oxygen and moisture of the air. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Carbon compounds have low melting and boiling points. Reason (R): Intermolecular forces in covalent compounds are weak. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. 12 g of carbon burns completely in 32 g of oxygen. The mass of carbon dioxide formed is: 1

(a) 44 g **(b)** 28 g **(c)** 12 g **(d)** 20 g

6. Equal volumes of HCl and NaOH solutions of the same concentration are mixed. The resulting solution has pH: 1

(a) 7 **(b)** 13 **(c)** 1 **(d)** 3

7. Metal X reacts with dilute HCl giving H_2 but does not react with cold or hot water. Metal Y reacts with cold water. Which is more reactive? 1

(a) cannot be said **(b)** X **(c)** both equally **(d)** Y

8. The balanced equation for the complete combustion of methane is: 1

(a) $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ **(b)** $CH_4 + 3\,O_2 \longrightarrow CO_2 + 2\,H_2O$ **(c)** $2\,CH_4 + O_2 \longrightarrow 2\,CO_2 + H_2O$ **(d)** $CH_4 + O_2 \longrightarrow CO_2 + H_2O$

9. A metal X turns black on heating in air. The black product, on passing hydrogen over it, turns reddish brown. X is: 1

(a) zinc **(b)** copper **(c)** magnesium **(d)** iron

10. A white powder X on adding water forms a hard solid mass. X is: 1

(a) plaster of Paris **(b)** bleaching powder **(c)** washing soda **(d)** baking soda

11. A student heats a green carbonate ore of copper. The black residue and gas are: 1

(a) Cu and CO_2 **(b)** Cu_2O and O_2
(c) CuO and CO_2 **(d)** CuO and CO

12. A hydrocarbon X decolourises bromine water and has a molecular mass of 28 u. X is: 1

(a) ethyne **(b)** ethene **(c)** methane **(d)** ethane

13. Assertion (A): Decomposition of calcium carbonate is an endothermic reaction. Reason (R): Heat energy is absorbed to break the bonds in the compound. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. Assertion (A): Acid should always be added to water and not water to acid. Reason (R): Dilution of acid is a highly exothermic process. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. Assertion (A): Aluminium is a highly reactive metal, yet it is used to make cooking utensils. Reason (R): Aluminium forms a thin, impervious oxide layer on its surface. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. 12 g of carbon burns completely in 32 g of oxygen. Find the mass of carbon dioxide formed and name the law applied. 2

17. When a few drops of conc. HCl are put on a piece of limestone, bubbles form. Name the gas, test and equation. 2

18. A metal X reacts with dilute HCl and also with NaOH producing hydrogen. Identify the nature of X's oxide with an example metal. 2

19. Write the balanced equation for the complete combustion of ethane and name the products. 2

20. Why is the mass of the residue less than the mass of lead nitrate taken when it is heated in an open tube? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Balance $\mathrm{PbO} + \mathrm{C} \longrightarrow \mathrm{Pb} + \mathrm{CO}$ and identify the oxidising and reducing agents. 3

22. A student is given solutions of pH 1, 4, 7, 10 and 13. Which will turn phenolphthalein pink and which can neutralise a base? Arrange by acidity. 3

23. In the reaction $\mathrm{ZnO} + \mathrm{C} \longrightarrow \mathrm{Zn} + \mathrm{CO}$ identify the oxidising agent, reducing agent and the type of reaction. Why is it used for zinc? 3

24. Write the structural formula of propene and state two ways in which it differs from propane. 3

OR

Green coating forms on copper vessels kept in humid air. Name the coating, give its cause and one prevention. 3

25. A white solid X on heating gives $\mathrm{CO_2}$ and a residue Y. Y dissolves in water, is basic and is used in whitewash. Identify X and Y and write equations. 3

OR

Why does iron rust faster near the sea? Explain the role of salt in rusting. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Three test tubes contain iron nails: A with dry air (anhydrous calcium chloride), B with boiled water and oil, C with tap water open to air. State the result in each, the conclusion and one method of prevention. 5

OR

Compare the properties of HCl and $\mathrm{CH_3COOH}$ of the same concentration with respect to pH, conductivity and reaction with Mg. Explain. 5

27. Four metals A, B, C and D: A displaces B, C and D from their salt solutions; B displaces C and D; C displaces D; D displaces none. Arrange them in order of reactivity and suggest a suitable extraction method for A and for D. Name one possible set of four metals. 5

OR

Explain conversion of ethanol to ethene, ethanoic acid and ethyl ethanoate with reagents and equations. Identify the oxidising agent and the catalyst. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND